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Performance of polynomial Gaussian functions in describing the molecular electronic continuum

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Abstract. The continuum molecular orbitals arising from L^2 calculations using special polynomial Gaussian functions are accurately tested to probe their capability in producing accurate photoemission cross sections. The molecular potential is represented by the local Xα approximation, which permits a comparison with numerically integrated (exact) orbitals. Several strong anisotropic molecules are taken into account to check the flexibility of the L^2 bases in the presence of strong attractive potentials. This work shows that for adequate choices of Gaussian functions, phase shifts may be estimated with reasonable accuracy while the errors in the continuum transition probabilities are within 2–3% in the more unfavourable cases. Therefore, the proposed basis sets appear to be suitable for the description of the electronic continuum in molecular photoionization calculations.

1. Introduction

Square-integrable L^2 functions have been used extensively in computations requiring the description of the electronic continuum of molecules. Despite incorrect asymptotic behaviour, with an adequate choice of basis set, accurate molecular results were obtained for integral [1–5] and differential [6–8] photoemission cross sections and for autoionizing line widths [9, 10]. The reasons for such a success arise from the need to represent the orbitals lying in the continuum in only a relatively small region around the molecule. The problem inherent with the incorrect damping at large r may be adequately overcome with special *ad hoc* techniques. For instance, if one is interested in the total cross section only, the Stieltjes–Tchebycheff moment theory (STMT) provides a powerful tool to recover the continuum normalization from a set of 'discretized' excited states [11–14]. The evaluation of partial differential cross sections requires additional information such as the density of states and the asymptotic phase shifts, that must be extracted from the L^2 continuum orbitals [5, 15]. These are generally obtained by a matching with the corresponding regular and irregular Coulomb wave in a region far from the molecule where the potential is purely Coulombic. The several types of L^2 bases employed to represent the molecular electronic continuum have been described and compared by Fortunelli and Carravetta [16].

In this family there are the polynomial Gaussian-type orbitals (PGTO), initially proposed by Kaufmann *et al* [17] and subsequently employed for atoms and for molecules having only one heavy atom [9, 18, 19]. The main advantage of these bases comes from the analyticity of the multi-centre integrals which allows one to expand the bound orbitals in a multi-centre basis and the continuum ones in a fixed unique centre. This permits one to avoid expensive one-centre expansions of the bound molecular orbitals as well as of the Coulomb and exchange potentials which are needed by other methods [20, 21]. Moreover, when suitable techniques such as the K-matrix adapted for the L^2 basis [22, 23] are employed to recover the electronic continuum degeneration, multi-centre localized functions can be added to improve the convergence of the partial-wave expansion [19].

However, no systematic test covering a wide range of energies for high anisotropic molecules has been reported in the literature. In this paper we will show that with a suitable choice of the PGTOs, accurate results may be obtained both for the asymptotic phase shifts and for transition moments. The calculations are carried out on molecules with more 'heavy' atoms which originate strong attractive potentials very different from the hydrogenic or atomic cases. The adopted Xα [24] approximation furnishes very realistic one-electron local potentials which allow a comparison between the L^2 results and 'exact' results from numerical integration of the resulting Schrödinger equation. Great importance is given to the redundancy of the basis set, which may cause loss of numerical precision and which prevents one from extending arbitrarily the number of functions.

In the next section we present the method and the basis sets considered. Then we discuss the results and finally we propose a small modification to the computational scheme currently employed in our group to treat photoionization processes in molecules.

2. Method and computational details

Starting from a molecular N-electron closed-shell SCF state, a local approximation of the operator describing single excitations from the double occupied orbital j is

$$\hat{h}_j^{N-1} = \hat{T} + \hat{V}_{\mathrm{ne}} + \sum_{i=1}^{N/2}(2 - \delta_{ij})\hat{J}_i - 3\alpha(3\rho/8\pi)^{1/3} \tag{1}$$

where $\hat{T}$ is the kinetic energy operator, $\hat{V}_{\mathrm{ne}}$ the nuclear attraction potential, $\hat{J}$ the two-electron Coulomb operator and ρ the α (or β) spin density of the SCF ground state. The exchange terms have been replaced by the Xα local potential which has been widely used in photoionization studies [3, 25, 26] and has proven to be an approximate but very realistic model.

For waves whose angular behaviour is described by real spherical harmonics $S_{lm}(\theta, \varphi)$, the one-electron radial eigenvalue equation is

$$\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{l(l+1)}{2r^2} + V_{lmj}(r)\right]\phi_\mu(r) = \epsilon_\mu\phi_\mu(r) \tag{2}$$

$$V_{lmj}(r) = \int \sin\theta\,\mathrm{d}\theta \int \mathrm{d}\varphi\, V_j(r, \hat{r})\,|S_{lm}(\theta, \varphi)|^2 \tag{3}$$

where $V_j(r, \hat{r})$ includes all the potential terms of equation (1). The angular integration is carried out numerically for a large number of r values and the resulting function is then interpolated with cubic splines. A numerical integration of equation (2) on a suitable mesh of points using standard techniques [27] may be performed and the results are practically exact.

With the aim to test the capability of L^2 functions to represent the electronic continuum, several basis sets are considered. The first one, labelled $g0$, is formed by simple GTO spherical functions:

$$g0: \quad f_j(r) = N_j r^l \mathrm{e}^{-\alpha_j r^2} \tag{4}$$

where l is the azimuthal quantum number and N_j is the normalization constant. The exponents form a geometrical sequence: $\alpha_{j+1} = \eta\alpha_j$, where η is a constant mainly chosen on the basis of a redundancy criterion. For this and for the following basis sets, the angular part of the functions is described by real spherical harmonics. Another family of functions is

$$gn: \quad f_j(r) = N_j r^{l+n} \mathrm{e}^{-\alpha_j r^2} \tag{5}$$

with α_j in geometrical sequence and with n an even integer. These sets have to be supplemented by a small number of simple GTOs with high exponents for a better description of the wavefunction near the origin.

Two other families of functions are implemented:

$$pn: \quad f_j(r) = N_j r^l \mathrm{e}^{-\alpha_j r^2}(r^n - a_j^n) \tag{6}$$

$$pnm: \quad f_j(r) = N_j r^l \mathrm{e}^{-\alpha_j r^2}(r^n - a_j^n)(r^m - b_j^m) \tag{7}$$

where a_j and b_j are positive constants and again a few simple GTOs are added. Although any even power could be considered, in this work only the $n = m = 2$ case is taken into account. For the $p2$ and $p22$ sets the zeros of the polynomials a_j form a geometrical sequence $a_{j+1} = \tau a_j$ from $\simeq$1 to $\simeq$50 au. In $p2$ each α_j is computed by imposing that the two maxima of the absolute value of the function $g_j = rf_j$ have a definite ratio, i.e. $|g_j(r_2)/g_j(r_1)| = K$ where r_2 and r_1 $(< r_2)$ are the radial coordinate values where $\mathrm{d}g_j/\mathrm{d}r = 0$. For $p22$ both α_j and b_j are chosen to satisfy a similar constraint, i.e. $|g_j(r_3)/g_j(r_2)| = |g_j(r_2)/g_j(r_1)| = K$, r_3, r_2 and r_1 are the extrema of the g_j function $(r_1 < a_j < r_2 < b_j < r_3)$. Hence in the $p2$ and $p22$ the a_j sequence is assigned, and the remaining parameters α_j and b_j are determined according to K, the optimal value of which was seen to be $\simeq$1.5.

Equation (2) is projected on these basis sets and the resulting matrix eigenvalue problem is solved by standard diagonalization techniques. The resulting projected eigensolutions and related quantities may be compared with those obtained by numerical integration on a radial grid. This furnishes a useful test on the performances of the L^2 basis sets and it will be illustrated in the next section.

To employ these basis sets in molecular calculations it is important to control their redundancy. For all the basis sets considered, the lowest eigenvalue of the overlap matrix was always $>10^{-5}$. Although it was seen that more redundant bases are sometimes effective in yielding accurate results, these are not considered because their use in molecular calculations could be dangerous and lead to unpredictable results. From the redundancy point of view a comparison of the several bases reveals some interesting features. It is seen that in going from $g0$ to $g2$ and to $g4$ the crucial parameter η may be considerably decreased. This stems from the stronger variation versus r of the single functions as increasing the power of r. Thus more functions spanning the same r region can be included and, in principle, a greater flexibility of the higher power sets is expected.

During the optimization of the basis sets it was observed that the inclusion of very diffuse functions only has the effect of increasing the number of high Rydberg orbitals with no appreciable improvement of the continuum orbitals. Since in effective calculations the discrete excited orbitals may be conveniently described by multi-centre basis sets, we concentrate here only in the continuum spectral region. On the other hand, a good description of the discrete orbitals is an important prerequisite to obtaining good continuum orbitals. Indeed, in variational calculations the orthogonality of the eigensolutions does not allow one, in principle, to obtain an accurate eigenfunction without describing the lower ones at least at the same level of accuracy. Thus the orbitals in the continuum should be orthogonal to the whole Rydberg series. However, for practical use of L^2 expansions, it suffices assuring an accurate description only of those Rydberg orbitals differing from each other in the r region spanned by the L^2 continuum functions. In the present case we manage the basis set in such a way to represent up to 7–8 Rydberg orbitals, this number being adequate to realize a correct orthonormalization up to $r = 40$–50 bohr from the expansion centre.

As noticed by Kaufmann *et al* [28] the success of an L^2 basis set in representing orbitals in the continuum region strongly requires a proper choice of the single functions. We confirm this conclusion by underlining the need for a very 'regular' sequence of functions to avoid unphysical behaviours which are strongly dependent on the basis set. By 'regular' we mean that the α_j sequence must obey a defined algebraic law in which each element is connected with the other ones. The geometrical sequence adopted here both for the g and for the p bases, furnishes an example of this and has the relevant advantage of yielding an overlap matrix with $\langle f_j | f_{j+1} \rangle \simeq$ constant. With this in mind, particular care has to be taken in connecting the sequence of standard GTOs (for the short-range description) with the sequence of special polynomial GTOs. The overlap between the more expanded GTO and the more contracted PGTO play an important role in assuring a balanced description of the orbitals in going from the near to the asymptotic radial region.

3. Results and discussion

In this work we consider the N_2 and CO_2 molecules with excitations from the outer valence orbitals $3\sigma_g$ and $3\sigma_u$, respectively. Experimental geometries are used: $R(\mathrm{NN}) = 2.0673$ bohr and $R(\mathrm{CO}) = 2.192$ bohr. The SCF calculation was carried out using standard 6-31G* [29] bases for both molecules. The wave expansion centre was always placed on the molecular barycentre and the off-centre nuclei were placed on the z-axis.

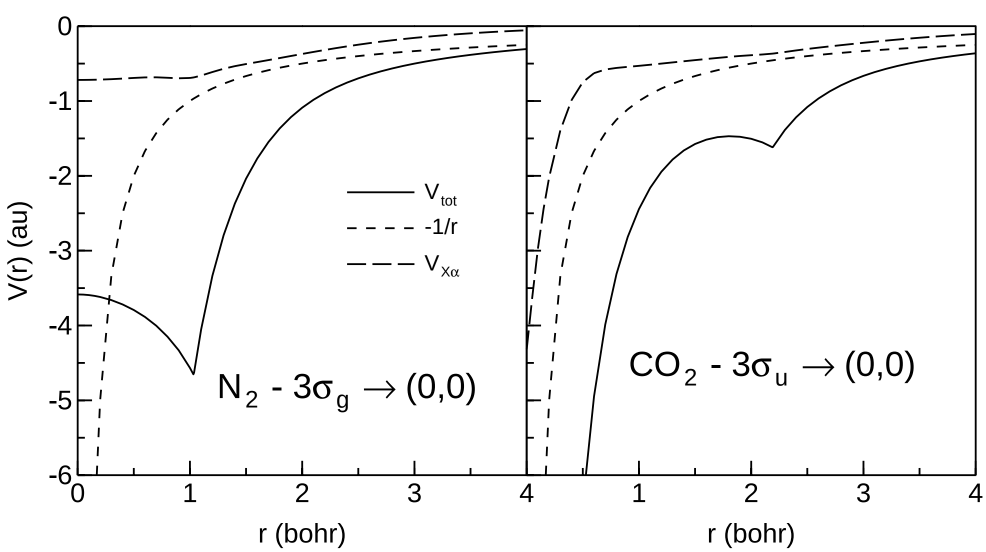

Figure 1. Radial potentials for s waves for N_2 and CO_2 arising from the expulsion of one electron from an SCF occupied orbital. The Xα contribution and the pure Coulombic potential are also displayed as broken curves.

3.1. s waves

The s waves are by far the most difficult to represent since the absence of the centrifugal potential causes many states to fall in the discrete energy region and it is hard to obtain high densities of states in the continuum. Moreover, they have appreciable values even near the origin, where the nuclei of the molecule yield potentials with high radial derivatives. These unfavourable features provide a highly demanding test of the capability of the basis sets with $l = 0$.

The potentials originating from the mentioned molecules, computed by equation (3) with $l = 0$, $m = 0$, $j = 3\sigma_g$ for N_2 and $j = 3\sigma_u$ for CO_2, are reported in figure 1. The pure Coulombic potential and the local exchange term are also displayed. The molecular potentials appear to be very different from the pure Coulombic one. The potential that originated from CO_2 is more attractive at all radial distances and goes as $-6/r$ as $r \to 0$. The N_2 potential has no pole in the origin (where for symmetry constraints the radial derivative is null) and crosses the hydrogenic potential near $r = 0.5$ bohr. Both curves display a local minimum at a radial distance corresponding to the presence of one or more nuclei. In both cases the Xα contribution varies smoothly versus r and goes rapidly to zero outside the molecular region.

An example of s waves for these two molecules using the $g6$ basis is presented in figure 2. Together with an L^2 orbital the exact solution is reported for comparison. To appreciate the effect of the short-range molecular potential with respect to the hydrogenic case, the Coulombic regular function of the same energy is also reported. All functions are $\delta(E)$ normalized. The normalization of the L^2 orbitals was achieved by minimizing the absolute differences between the L^2 and the exact wave from the origin up to the second radial node. It appears that near the origin the molecular waves are 'compressed' by the strong attractive potential and a phase shift originates. This does not change appreciably outside the molecular region.

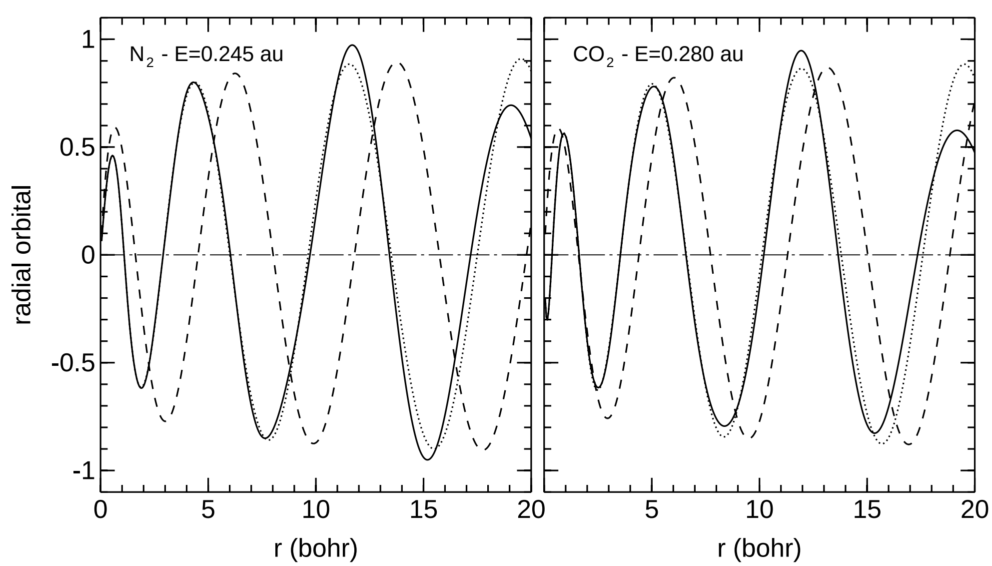

Figure 2. Continuum s orbitals for N_2 and CO_2 versus the radial distance. Full curves, L^2 orbitals; dotted curves, exact orbitals; broken curves, pure Coulombic orbitals with the same energy.

The L^2 orbitals are very close to the exact solutions until the first three to four radial oscillations, while the damping due to the exponential factor of the L^2 functions is clearly evident from 15 bohr (fifth–sixth radial node) onward. The same trend is roughly observed for the other L^2 orbitals. Thus the higher the wave energy, the smaller the radial region will be where the approximate orbitals match the exact ones.

While the $g4$, $g6$ and $p22$ bases yield orbitals of comparable accuracy, the $g0$, $g2$ and $p2$ bases are inferior. Moreover, the apparently best bases are capable of producing higher density of states. For example, the number of states with $E < 1$ au for N_2 was 5, 6, 7, 8, 6, 9 for $g0$, $g2$, $g4$, $g6$, $p2$, $p22$, respectively.

The performances of the several basis sets can be estimated in a more quantitative way by computing the asymptotic phase shift of the variational L^2 orbitals. This was done by matching each continuum L^2 orbital with a linear combination of the corresponding regular and irregular Coulomb waves outside the molecular region. It was noticed that for waves with energy up to $\simeq 0.8$ au it is possible to find a fitting range (for instance from 5–10 bohr) giving phase shifts practically independent of its extrema. In contrast, the greater the wave energy, the smaller is the region where the L^2 orbitals are reliable and the phase shift must be evaluated in r ranges closer and closer to the origin. On the other hand, to avoid an underestimation of the phase shift the fitting range cannot enter the molecular region. Thus a compromise between these two requirements is inevitable and the choice of the fitting range may affect the resulting phase shift. This unreliability is more and more evident as the energy increases.

The computed and exact phase shifts are displayed in figures 3 and 4 for N_2 and CO_2, respectively. For the first molecule good estimates are obtained up to $\simeq$1 au by all bases

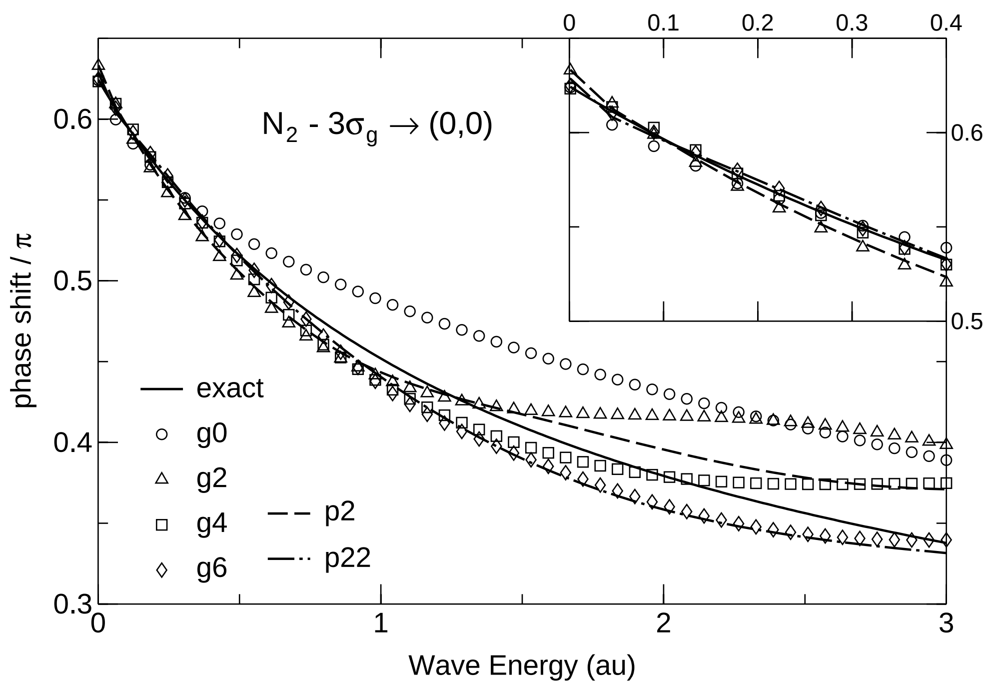

Figure 3. Phase shift of s waves arising from the N_2 potential of figure 1 for different basis sets. The inset shows a magnified view of the near-threshold energy region.

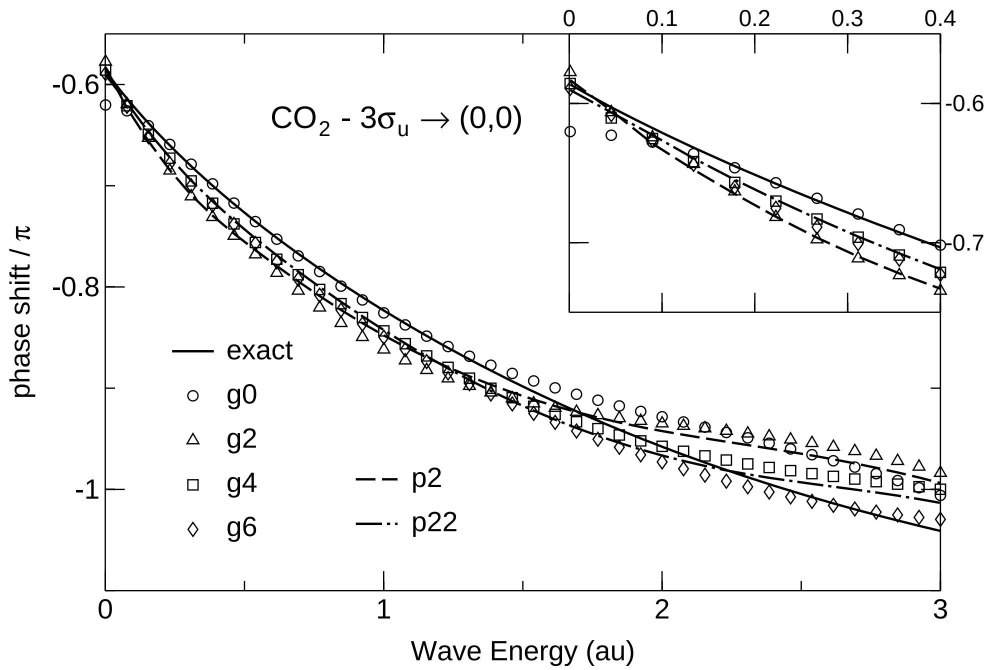

Figure 4. Phase shift of s waves arising from the CO_2 potential of figure 1 for different basis sets. The inset shows a magnified view of the near-threshold energy region.

except $g0$. The $g6$ and $p22$ bases give the best results; their maximum error is less than 0.08 rad at all energies, with minor errors in the low-energy region. According to the previous discussion it appears that for N_2 the energy range where accurate phase shifts are obtained increases monotonically with the power of the gn bases. For CO_2 the $g0$ results show some deviations from the exact value even near the threshold; $g6$ seems to yield more accurate results. It is worth noting that in the hydrogenic case much smaller errors are obtained. For instance, the $g6$ results show errors < 0.01 rad up to $E = 1$ au and < 0.02 rad up to $E = 3$ au. Thus it is evident that different performances of the basis set are required for hydrogenic and the present molecular potentials.

More encouraging results are obtained for the second quantitative test concerning the cross section. The initial state is the first discrete state with $l = 1$ originating from the same molecular potentials of figure 1 by adding the correct centrifugal term. Although this is an 'atomic' orbital, it is representative of the projection of typical molecular orbitals in the $l = 1$ space. This device also has the advantage of yielding dipole transition moments which, in the limit of exact eigenstates, satisfy the hypervirial relation. The initial state has $\langle r\rangle \simeq 1.0$ bohr and $E = -1.277$ au for N_2 and $\langle r\rangle \simeq 1.5$ bohr and $E = -0.881$ au for CO_2.

The exact cross section is reported in the upper part of figures 5 and 6, while the percentage errors resulting from L^2 calculations for both length and velocity gauges are displayed in the lower parts. This choice permits us to point out even small differences between the exact and approximate transition probabilities. Only the results given by the best basis sets are shown for clarity.

The results are clearly satisfactory; in all cases the percentage errors are less than 5%. The velocity transition moments seem to be the most reliable. This is not surprising, since

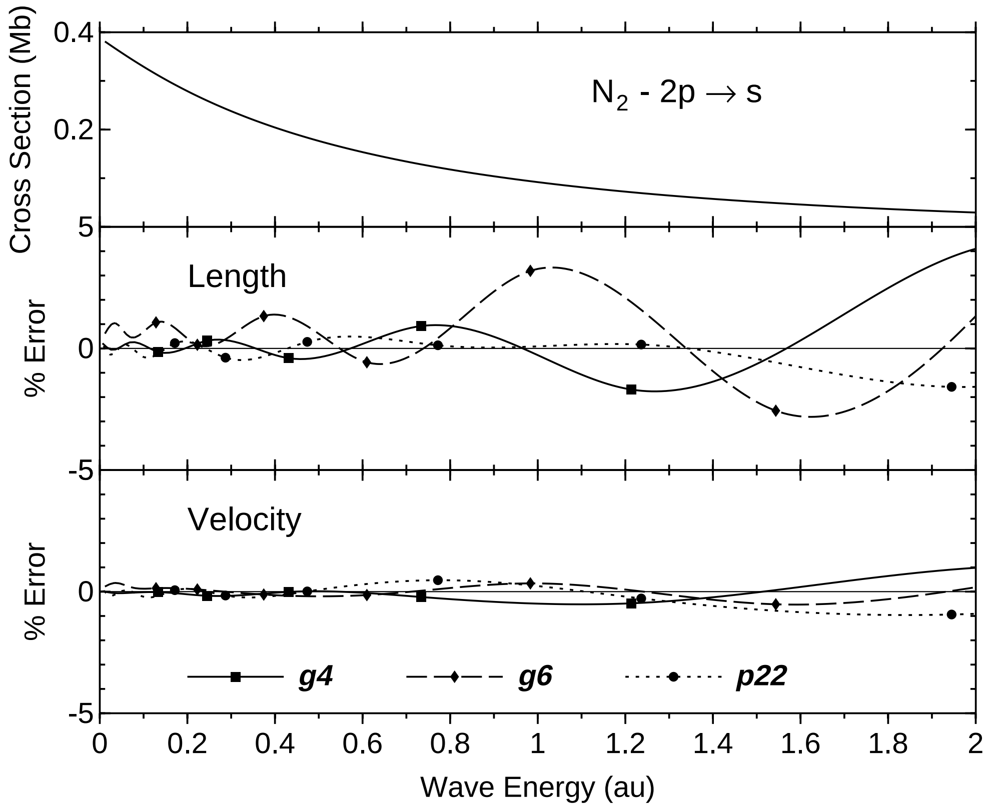

Figure 5. Cross section for the 2p → s transition in N_2. Upper part, exact cross section from numerical integration; central part, percentage errors in the cross section for different basis sets in the dipole length gauge; lower part, percentage errors in the cross section for different basis sets in the dipole velocity gauge. The symbols match the energy of the L^2 orbitals from 0.1 au onward.

in the dipole gradient integrals the contribution of the L^2 orbitals near the origin is more relevant in comparison with the dipole length operator. On the other hand, the radial derivative in the dipole velocity transition moments makes this result not easily predictable. Another gratifying feature is the oscillating behaviour around the zero line of the cross section errors; this indicates that the L^2 orbitals are not affected by systematic deviations from the exact solutions. The $p22$ set seems to be the more reliable, at least in the present cases, its percentage error never exceeding 2%. A further reason to prefer the $p22$ basis lies in the small gauge dependence of the cross section as it appears mainly for N_2. The comparison with the hydrogenic case again stresses the importance of the potential on the performances of a basis set designed to represent continuum orbitals. The cross section errors are indeed smaller for a factor of 4–5 more or less for all basis sets.

Finally, we mention the possibility of increasing the degree of the polynomial in the pnm bases. A calculation using a $p44$ basis set gives much more accurate results. The errors were < 0.02 rad and $< 0.5\%$ for the phase shift and for the cross section up to $E = 4$ au, respectively. However, we consider the $p44$ basis with some caution because in effective molecular calculations their use may be discouraged both for the high computational cost

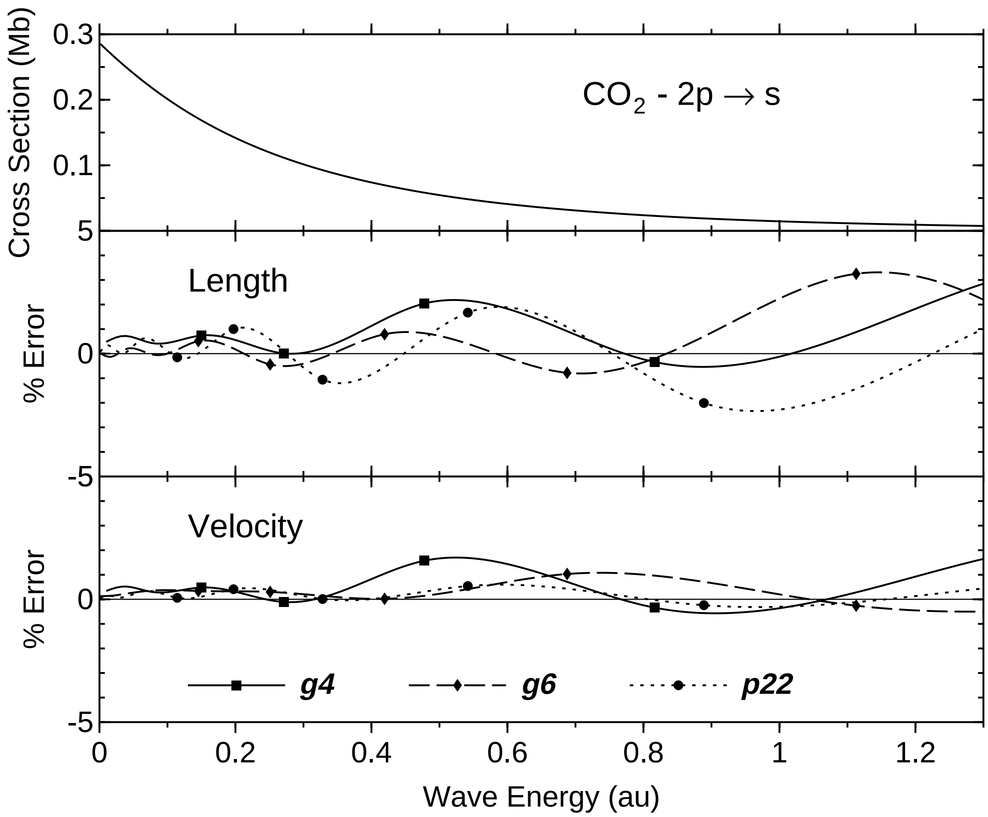

Figure 6. Cross section for the 2p → s transition in CO_2. Upper part, exact cross section from numerical integration; central part, percentage errors in the cross section for different basis sets in the dipole length gauge; lower part, percentage errors in the cross section for different basis sets in the dipole velocity gauge. The symbols match the energy of the L^2 orbitals from 0.1 au onward.

of the related two-electron integrals and for the very high ($\simeq 10^{12}$) polynomial coefficients of opposite sign which may affect the precision of the molecular integrals.

3.2. p waves

In comparison to the s waves, the p waves, in general, show a higher density of states in the continuum even for highly attractive potentials. Thus more accurate results are expected. Some relevant results of the $(1, 0)$ waves arising from the $N_2^+(3\sigma_g)^{-1}$ potential are displayed in figure 7. The orbitals shown in the upper left-hand part of the figure demonstrate the marked differences between the molecular and the pure Coulombic case. One extra radial lobe is found near the origin where a well in the potential is found. As for the s potentials, the Xα term is smooth and the strong attractiveness arises mainly from the electrostatic terms. The reported orbital is clearly a very good representation of the numerically integrated one; small differences appear from radial distances greater than 4 bohr. The L^2 phase shifts (upper right-hand part) closely follow the exact value. Some differences become evident as the energy increases for the $g4$ and $p22$ bases while larger errors are observed for the $g0$ basis. The best result ($p22$) gives absolute errors everywhere less than 0.070 rad. The cross

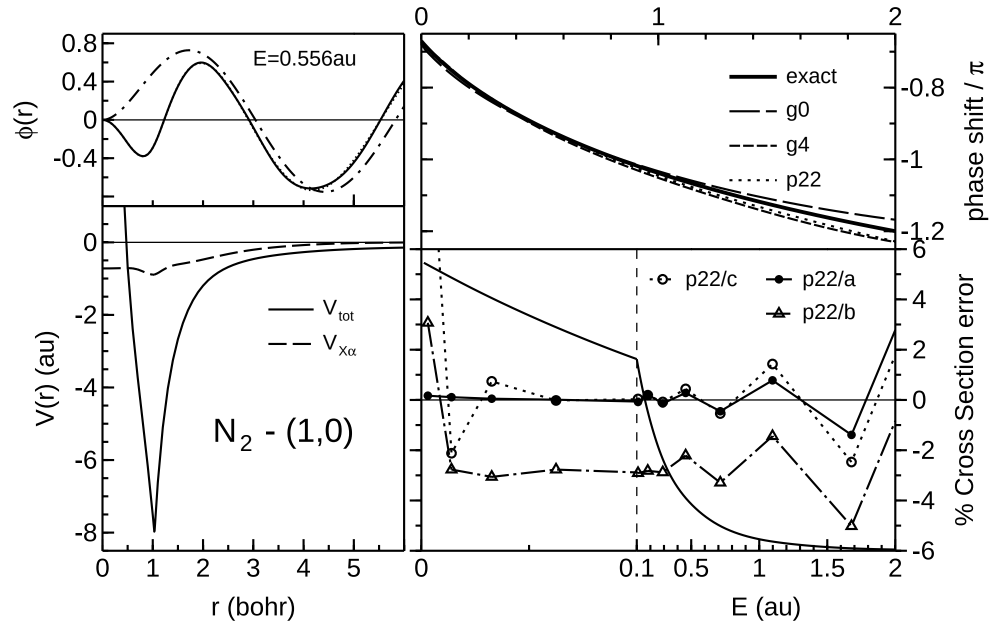

Figure 7. Relevant results of the (1, 0) waves arising from the $N_2(3\sigma_g)^{-1}$ core charge. Lower left-hand part, radial potential for (1, 0) orbitals; upper left-hand part, waves of energy 0.556 au; full curve, L^2 orbital; dotted curve, exact orbital; chain curve, regular Coulombic orbital; upper right-hand part, phase shift versus wave energy; lower right-hand part, cross section versus wave energy (the 0–0.1 energy range is strongly expanded). The full curve without symbols is the cross section in arbitrary units from the 2s orbital; the other curves with symbols refer to three different ways of achieving the continuum density normalization for the $p22$ basis: (a) minimization of the absolute differences between the L^2 and the exact waves; (b) $C_j = [2/(E_{j+1} - E_{j-1})]^{1/2}$; ($c$) the method proposed in [9].

section data (lower right-hand part) point out the high accuracy of the transition moments taken from the 2s initial state originating from the same potential without the centrifugal term. Errors greater than 1% are only found for energies above 1 au, where the cross section practically vanishes. The quality of the results of the other basis sets is in line with those already presented for the s waves.

The results of other methods for the estimate of the density of states in the continuum are reported in figure 7. The commonly used $C_j = [2/(E_{j+1} - E_{j-1})]^{1/2}$ underestimates this density with errors in the cross section of about 3–4%. Moreover, abrupt changes may be found near the threshold where there is an ambiguity in the determination of the normalization constant of the first continuum state. The more sophisticated algorithm of Macías *et al* [9] works in the full energy spectrum, but for low wave energies it sometimes leads to unpredictable results. These methods are strictly based on the ‘regularity’ of the discretized energies in the continuum, which is broken in passing through the continuum threshold.

All in all it appears that the present evaluation of the density of states in the continuum leads to a greater reliability of the results with respect to methods based on the density of ‘discretized’ orbitals in the continuum. Since this trend is generally observed for the other cases considered here, it appears that knowledge of the exact orbitals can be used to improve the quality of the L^2 transition moments considerably.

3.3. *f waves*

The last test reported here is summarized in figure 8 and refers to the representation of the (3, 0) waves in N_2, arising from the promotion of one electron from the $3\sigma_g$ orbital. This case was selected since it is well known [30] that a shape resonance originates and a broad maximum in the photoionization probability is observed [31]. The small local maximum in the potential around 2–3 bohr is consistent with these features. Again the deep potential minimum at 1 bohr induces a corresponding absolute maximum in the continuum wavefunctions (upper left-hand part of figure 8) with no counterpart in the regular Coulombic function. The phase shift now shows a non-monotonic behaviour with a maximum for wave energy of $\simeq 0.4$ au. The three L^2 results reproduce the exact curve well; the more accurate basis set is apparently the $g2$ one which gives errors of less than 0.03 rad. The high accuracy of these f waves is also evident in the lower right-hand part of figure 8, where the cross section for the 3d $\rightarrow$ f transition is reported. The best results are those arising from the $p2$ basis set whose error never exceeds 1.2%, even for very small cross section. Both the $g0$ and $g2$ results are, however, of similar quality. The figure allows one to appreciate the high density of states in the continuum, which is crucial in describing quantities which vary non-monotonically with energy.

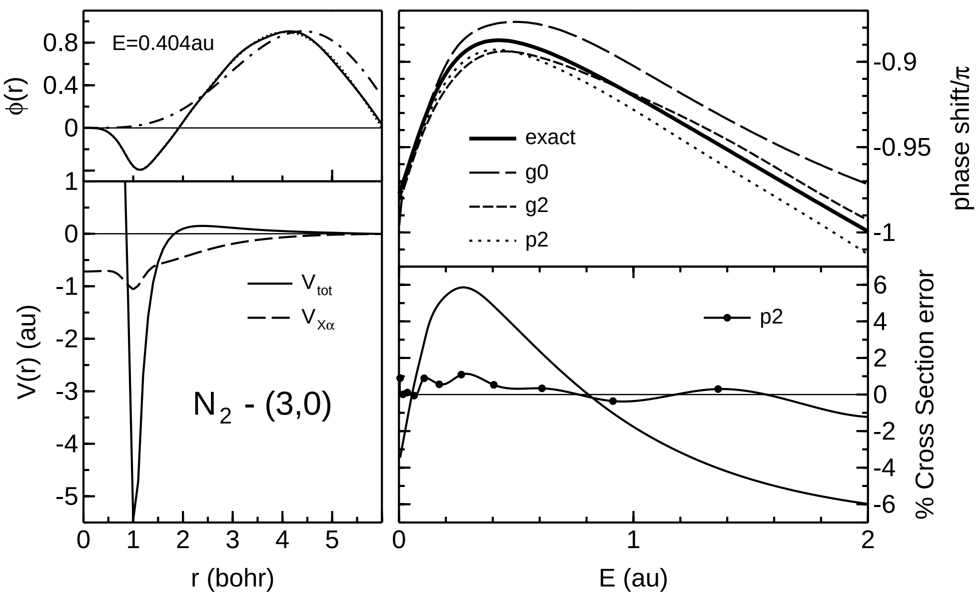

Figure 8. As in figure 7 for the (3, 0) waves. The cross section refers to the 3d orbital originated by the same potential with the right centrifugal term.

4. Conclusions

In this work we have presented a detailed study of the accuracy of L^2 orbitals in the continuum energy spectrum, the orbitals of which are represented by Gaussian functions eventually multiplied by special polynomials. The few cases reported here were selected with the aim of testing the flexibility of the several basis sets for radial potentials which are different from the pure Coulombic potential. Tests were also carried out for the C_2H_2, CO and H_2CO molecules as well as for the potentials arising from different cores of N_2 and CO_2. These results are in line with those presented here and do not deserve any particular comment.

The variety of cases considered here and others not presented for brevity leads to the following conclusions.

(i) With a proper choice of the L^2 functions accurate transition moments can be obtained. Reliable values of the $\delta(E)$ normalization constants C_j are required. The comparison with exact orbitals, easily available for local one-dimensional Hamiltonians, furnishes by far the more accurate estimates.

(ii) The asymptotic phase shift $\Delta(E)$ is accurate for low and medium energies (<1 au), while no stable algorithms were found for high wave energies. Moreover, the largest error was found for s waves, which exhibit the smaller density of states in the continuum.

These functions have already been employed in effective calculations of molecular differential photoionization cross sections. The results for H_2 [18], LiH and Li_2 [19] and N_2 [32] obtained without comparing the L^2 orbitals with the exact ones and using a special K-matrix technique to evaluate the phase shifts [18], are in good agreement with experiment and with other theoretical studies. The computational scheme is based on a definition of a set of zero-order partial-wave channels (PWC) followed by a sort of configuration interaction in the continuum using a K-matrix algorithm adapted for L^2 states [22]. The RPA implementation was also carried out [23]. An attractive feature of this method is the analytic evaluation of the one- and two-electron integrals without resorting to expensive one-centre expansions of the core orbitals and Fock-like potentials.

Despite such a success, it appears from this work that the possibility of comparing the L^2 orbitals with the exact continuum orbitals can enforce the reliability of the final results. This can be exploited by a simple modification of the zero-order states which were previously computed by non-local static-exchange Hamiltonians projected on spherical harmonics [22]. If these operators are replaced by suitable local Hamiltonians such a comparison will be possible and the two asymptotic quantities of interest can be properly evaluated with a modest increase of the computational cost. It is obvious that the true Hamiltonian operator must be preserved in the final K-matrix algorithm, which in our implementation works with general zero-order states [22]. With this device we think that this method becomes largely competitive with those employed by other groups [7, 15, 21, 33, 34].

Acknowledgments

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